

Monte Carlo Methods

Introduction

Generating Samples

Markov Chain Monte Carlo

Improving Efficiency

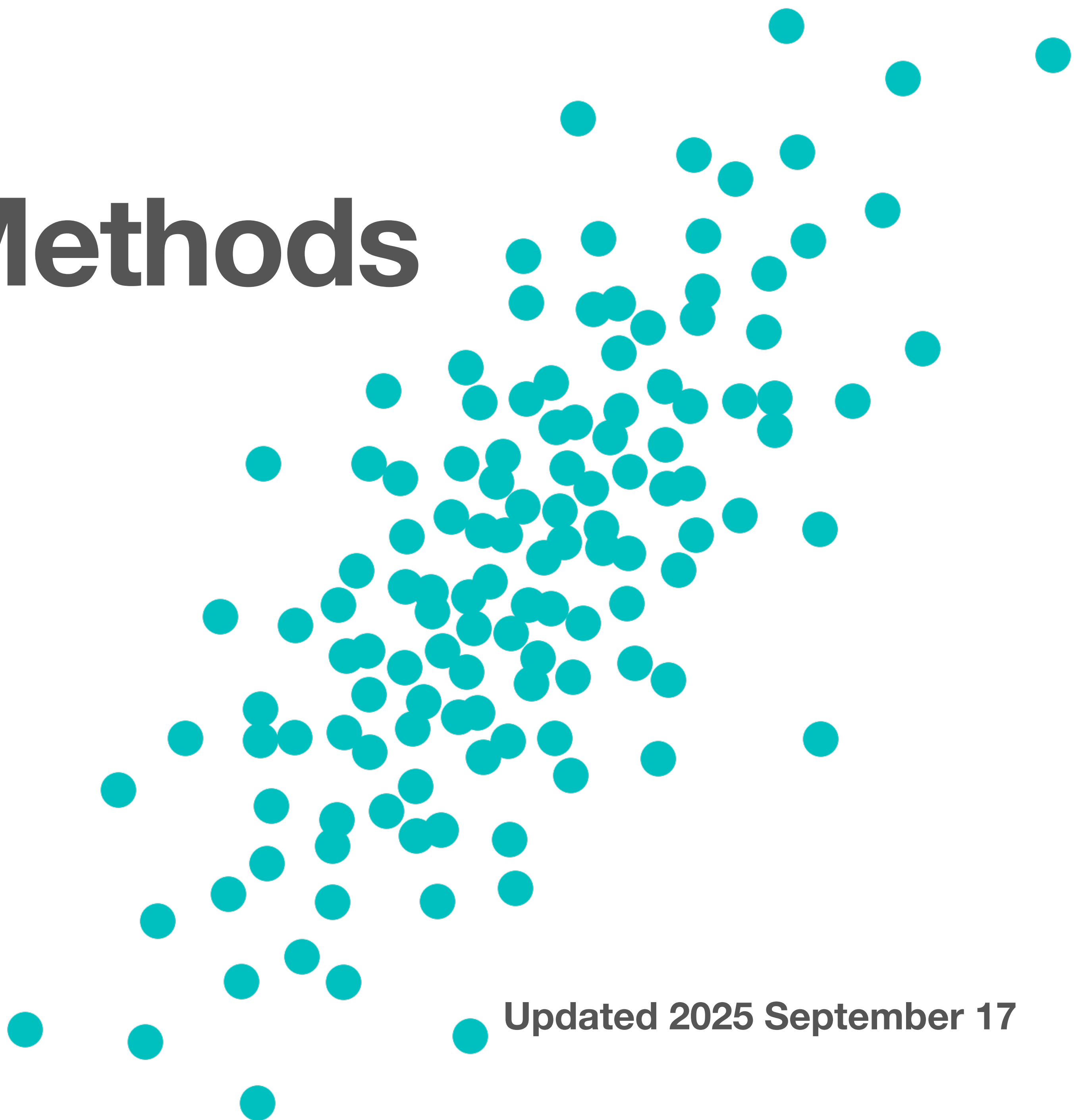
Selected Topics

Git website and repository

Canvas

Fred Hickernell, Fall 2025

Updated 2025 September 17



Generating Samples

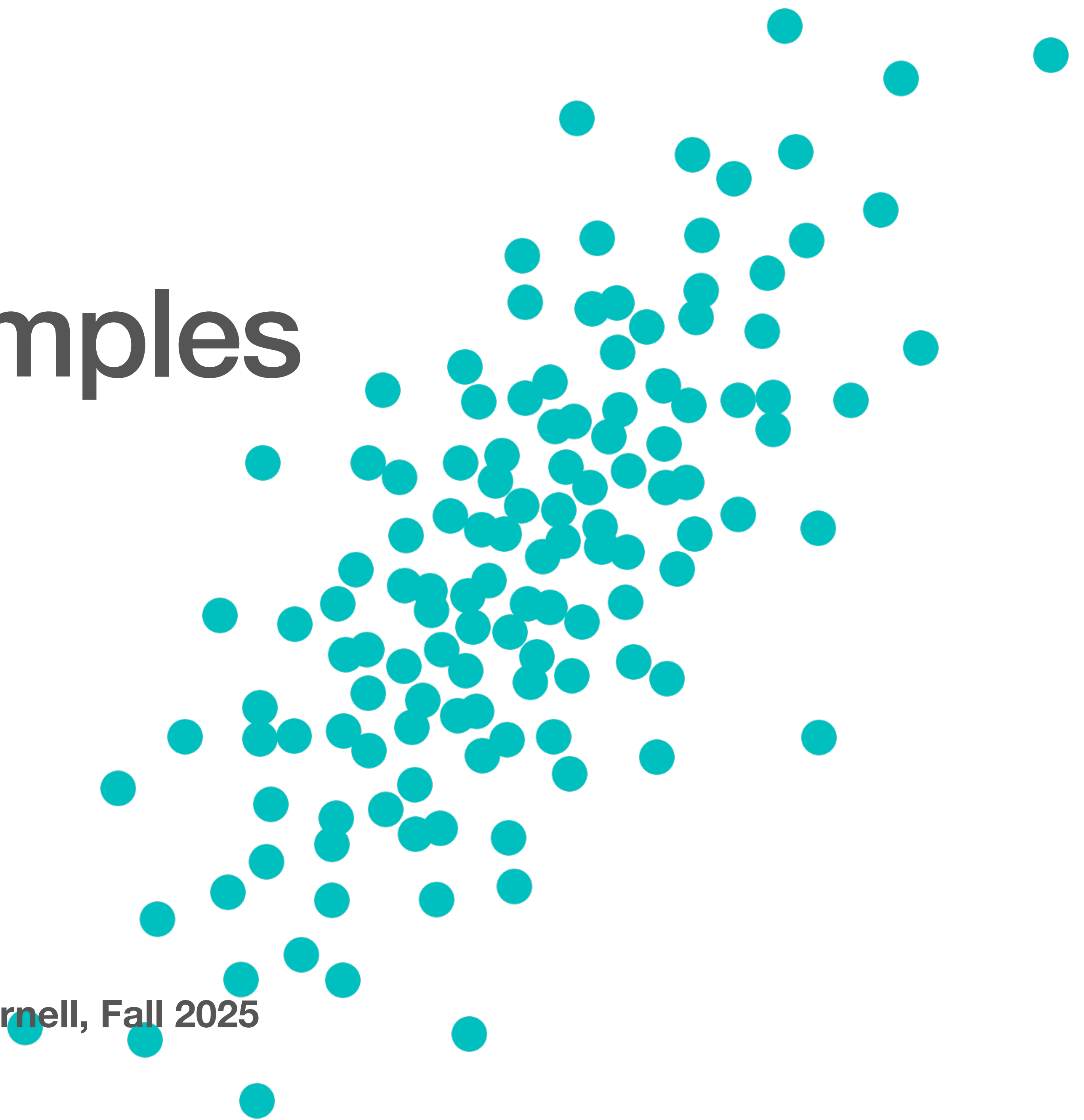
Owen, Chapters 3 – 6

Assignment 2 due Sep 19

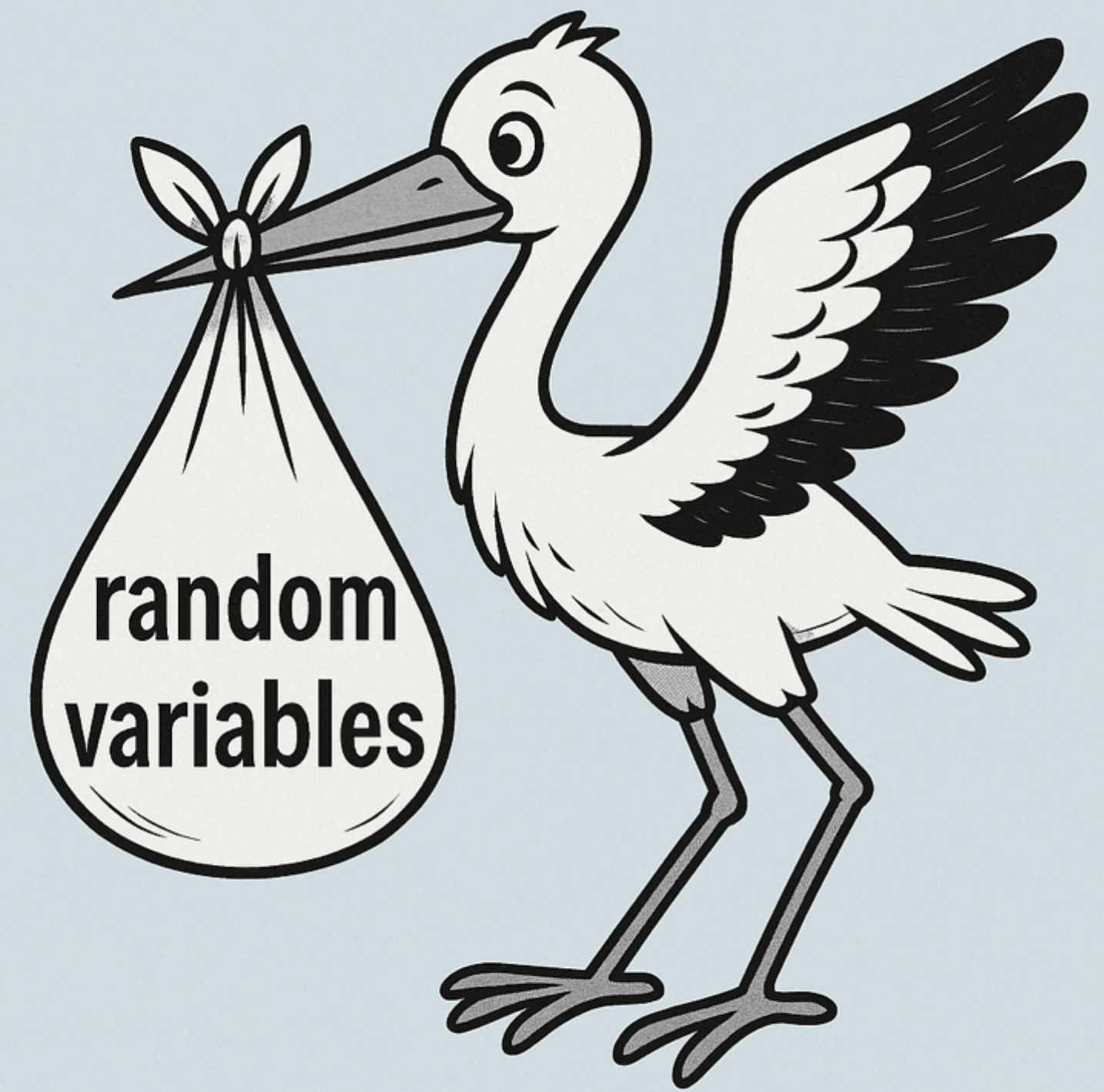
Test 1 Sep 24

Project Selection due Oct 1

MATH 565 Monte Carlo Methods, Fred Hickernell, Fall 2025



Where do random variables come from?





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- For a drastic failure see [RANDU](#)



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- Algorithms generate (typically **uniform pseudo-random** numbers
- Pseudo-random number generators are tested to ensure that their output **looks IID random** in every way possible
- Care must be taken when working on **multiple processors**
- For a drastic failure see [RANDU](#)
- **Physical** random number generators may be random, but cannot be guaranteed to be IID random of the desired distribution





Quantile function gives non-uniform random numbers

- Most random number generators output $U_1, U_2, \dots$ that mimic IID $\mathcal{U}[0,1]$.
- To get $X_1, X_2, \dots$ IID with cumulative distribution function F , we use the quantile function, Q , where $Q(u) := \inf\{x \in \mathcal{X} : F(x) \geq u\}$. Note that

$$Q(u) \leq x \iff u \leq F(x)$$

- Letting $X := Q(U)$, it follows that

$$\mathbb{P}(X \leq x) = \mathbb{P}(Q(U) \leq x) = \mathbb{P}(U \leq F(x)) = F(x)$$

and so $X \sim F$, and

$$X_1 := Q(U_1), X_2 := Q(U_2), \dots \stackrel{\text{IID}}{\sim} F$$



Binomial random numbers



Binomial random numbers

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Binomial random numbers

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x	0	1	2	3
PMF $\varrho(x)$	0.4^3 0.064	$3(0.6)(0.4^2)$ 0.288	$3(0.6^2)(0.4)$ 0.432	0.6^3 0.216
$x \in$	$[0, 1)$	$[1, 2)$	$[2, 3)$	$[3, \infty)$
CDF $F(x)$	0.064	0.352	0.784	1
$u \in$	$(0, 0.064]$	$(0.064, 0.352]$	$(0.352, 0.784]$	$(0.784, 1]$
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Note that F is **right**-continuous and Q is **left**-continuous



Explorations in generating random vectors

Generating Samples



Zero-inflated exponential

Suppose that X is a non-negative random variable with probability p_0 of being zero and otherwise an exponential distribution

$$F(x) = \begin{cases} 0 & -\infty < x < 0 \\ p_0 + (1 - p_0)[1 - \exp(-\lambda x)] & 0 \leq x < \infty \end{cases}$$

To generate $X_1, X_2, \dots \stackrel{\text{IID}}{\sim} F$ from $U_1, U_2, \dots \stackrel{\text{IID}}{\sim} \mathcal{U}[0,1]$, we need the **quantile function**



Quantile function for zero-inflated exponential

$$F(x) = \begin{cases} 0 & -\infty < x < 0 \\ p_0 + (1 - p_0)[1 - \exp(-\lambda x)] & 0 \leq x < \infty \end{cases}$$

$$\left. \begin{array}{l} u \leq F(x) \\ u = F(x) \end{array} \right\} \Rightarrow \begin{cases} x = 0, & 0 < u \leq p_0 \\ u = p_0 + (1 - p_0)[1 - \exp(-\lambda x)], & p_0 \leq u < 1 \end{cases}$$

$$\Leftrightarrow \begin{cases} x = 0, & 0 < u \leq p_0 \\ \frac{u - p_0}{1 - p_0} = 1 - \exp(-\lambda x), & p_0 \leq u < 1 \end{cases}$$

$$\Leftrightarrow x = Q(u) = \begin{cases} 0, & 0 < u \leq p_0 \\ -\frac{1}{\lambda} \log \left(1 - \frac{u - p_0}{1 - p_0} \right), & p_0 \leq u < 1 \end{cases}$$



Random vectors with **independent** marginals

Suppose that $X = (X_1, \dots, X_d)$ is a random **vector** with independent marginals.
Then

$$X = (Q_1(U_1), \dots, Q_d(U_d)), \quad U \sim \mathcal{U}[0,1]^d$$

has the desired distribution, provided that $Q_1, \dots, Q_d$ are the corresponding **quantile functions**.

We express n samples of X as an $n \times d$ matrix or array,

$$X = \begin{pmatrix} X_{11} & X_{12} & \cdots & X_{1d} \\ \vdots & \vdots & & \vdots \\ X_{n1} & X_{n2} & \cdots & X_{nd} \end{pmatrix}, \text{ where } X_{ij} \text{ is the } j^{\text{th}} \text{ **component** of the } i^{\text{th}} \text{ **sample**}$$



What is your expertise in computing and probability?

- Go to [menti.com](https://www.menti.com)
- Use code 4845 0474



Multivariate normal distribution

If $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}_d, \mathbf{I}_d)$, i.e., a d -dimensional standard normal random variable, then $\mathbf{X} = \mathbf{AZ} + \mathbf{b}$ (thinking of $\mathbf{Z}$ as a column vector) has

$$\boldsymbol{\mu} = \mathbb{E}(\mathbf{X}) = \mathbf{b}$$

$$\begin{aligned}\Sigma &= \text{cov}(\mathbf{X}) = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top] \\ &= \mathbb{E}[\mathbf{AZZ}^\top \mathbf{A}^\top] = \mathbf{AA}^\top\end{aligned}$$

So $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$, where Σ is **symmetric** and **positive-definite**. So, give a desired $\boldsymbol{\mu}$ and Σ , one needs only to find $\mathbf{A}$ with $\Sigma = \mathbf{AA}^\top$, and then

$$\mathbf{X} = \mathbf{AZ} + \boldsymbol{\mu} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$$

Note that there are **multiple** ways to decompose $\Sigma = \mathbf{AA}^\top$.



Cholesky decomposition of Σ

We want to find a lower triangular matrix A , such that $\Sigma = AA^T$.

$$\begin{pmatrix} 3 & -1 \\ -1 & 1 \end{pmatrix} = \Sigma = AA^T = \begin{pmatrix} a_{11} & 0 \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} a_{11} & a_{21} \\ 0 & a_{22} \end{pmatrix}$$

$$= \begin{pmatrix} a_{11}^2 & a_{11}a_{21} \\ a_{11}a_{21} & a_{21}^2 + a_{22}^2 \end{pmatrix}$$

$$\implies a_{11} = \sqrt{3}$$

$$a_{21} = -1/a_{11} = -1/\sqrt{3}$$

$$a_{22} = \sqrt{1 - a_{21}^2} = \sqrt{2/3}$$

$$A = \begin{pmatrix} \sqrt{3} & 0 \\ -1/\sqrt{3} & \sqrt{2/3} \end{pmatrix}$$



Gaussian processes

- Random **functions** such that evaluating them at a finite number of points gives a random vector with a multivariate normal distribution
- The distribution is denoted $\mathcal{GP}(\mu, K)$, where μ is the **mean** function and K is the **covariance** kernel
- $g \sim \mathcal{GP}(\mu, K) \implies$
 - $\mathbb{E}[g(t)] = \mu(t)$
 - $\mathbb{E}[\{g(t) - \mu(t)\}\{g(x) - \mu(x)\}] = K(t, x)$



Brownian Motion

Special case of a Gaussian process

$$\mu(t) = 0 \quad \text{and} \quad K(t, x) = \min(t, x), \quad 0 \leq t$$

For $0 \leq t_1 < t_2 < \dots \leq t_d$,

$$X = (B(t_1), \dots, B(t_d))^{\top} \sim \mathcal{N}(\mathbf{0}, \Sigma)$$

where

$$\Sigma = \begin{pmatrix} t_1 & t_1 & \dots & t_1 \\ t_1 & t_2 & \dots & t_2 \\ \vdots & \vdots & \ddots & \vdots \\ t_1 & t_2 & \dots & t_d \end{pmatrix} = \begin{pmatrix} \sqrt{t_1} & 0 & \dots & 0 \\ \sqrt{t_1} & \sqrt{t_2 - t_1} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \sqrt{t_1} & \sqrt{t_2 - t_1} & \dots & \sqrt{t_d - t_{d-1}} \end{pmatrix} \begin{pmatrix} \sqrt{t_1} & \sqrt{t_1} & \dots & \sqrt{t_1} \\ 0 & \sqrt{t_2 - t_1} & \dots & \sqrt{t_2 - t_1} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sqrt{t_d - t_{d-1}} \end{pmatrix}$$



Brownian Motion

For this Cholesky decomposition of the covariance matrix

$$\Sigma = \begin{pmatrix} t_1 & t_1 & \cdots & t_1 \\ t_1 & t_2 & \cdots & t_2 \\ \vdots & \vdots & \ddots & \vdots \\ t_1 & t_2 & \cdots & t_d \end{pmatrix} = \begin{pmatrix} \sqrt{t_1} & 0 & \cdots & 0 \\ \sqrt{t_1} & \sqrt{t_2 - t_1} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \sqrt{t_1} & \sqrt{t_2 - t_1} & \cdots & \sqrt{t_d - t_{d-1}} \end{pmatrix} \begin{pmatrix} \sqrt{t_1} & \sqrt{t_1} & \cdots & \sqrt{t_1} \\ 0 & \sqrt{t_2 - t_1} & \cdots & \sqrt{t_2 - t_1} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sqrt{t_d - t_{d-1}} \end{pmatrix}$$

One may construct, IID random samples $X_i = (B(t_1), \dots, B(t_d))^T \stackrel{\text{IID}}{\sim} \mathcal{N}(\mathbf{0}, \Sigma)$ by

$$X_{i1} = \sqrt{t_1} Z_{i1}, \quad X_{i2} = X_{i1} + \sqrt{t_2 - t_1} Z_{i2}, \quad \dots, \quad X_{id} = X_{i,d-1} + \sqrt{t_d - t_{d-1}} Z_{id}$$

where $Z_i \stackrel{\text{IID}}{\sim} \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$



Stock/Asset Prices

Brownian motions are used in financial modeling. If

- S_0 is the initial price of a stock/asset
- r is the interest rate
- σ is the volatility

Then

$$S(t) = S_0 \exp\left((r - \sigma^2/2)t + \sigma B(t)\right)$$

is a **geometric Brownian motion** model of the **random** stock price



Option Payoffs and Prices

Give some model of asset prices, we may model the payoff of an option.

Let K be the **strike price** and T be the time to maturity. Examples of **discounted payoffs** are

European call	$\max(S(T) - K, 0) \exp(-rT)$
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European put	$\max(K - S(T), 0) \exp(-rT)$
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Asian Arithmetic mean call	$\max\left(\frac{1}{T} \int_0^T S(t) dt - K, 0\right) \exp(-rT)$
----------------------------	--

The fair price of an option is

$$\mu = \mathbb{E}[\text{payoff}]$$



Low discrepancy sampling

If we are willing to accept samples that are not IID, then we may get substantial gains in the convergence rate. We say that

$$U_1, \dots, U_n \stackrel{\text{LD}}{\sim} \mathcal{U}[0,1]^d$$

if the **empirical distribution** of $\{U_1, \dots, U_n\}$ is close to the uniform distribution. The measure of how far away these distributions are is the **discrepancy**, and it will be defined precisely later. Some examples are

- Digital sequences, especially Sobol' sequences
- Lattice sequences
- Halton sequences
- Kronecker sequences

These come in **deterministic** and **random** forms.



Low discrepancy sampling

- LD sequences are available through [scipy](#) and [qmcipy](#), which has been developed in our research group
- Rows correspond to samples and columns to coordinates
- See [Generating Samples](#) for empirical comparisons
- The theory of LD sampling will be touched on later in the semester



Bias–variance decomposition

- Let μ be any population quantity, e.g., the mean
- Let $\hat{\mu}$ be any estimator for μ , e.g., the sample mean
- Then the mean squared error of $\hat{\mu}$ is

$$\begin{aligned}\text{mse}(\hat{\mu}) &= \mathbb{E}[(\mu - \hat{\mu})^2] = \mathbb{E}[\{\mu - \mathbb{E}(\hat{\mu})\} + \{\mathbb{E}(\hat{\mu}) - \hat{\mu}\}]^2 \\ &= \mathbb{E}[\{\mu - \mathbb{E}(\hat{\mu})\}^2] + \mathbb{E}[\{\mathbb{E}(\hat{\mu}) - \hat{\mu}\}^2] + 2\mathbb{E}[\{\mu - \mathbb{E}(\hat{\mu})\}\{\mathbb{E}(\hat{\mu}) - \hat{\mu}\}] \\ &= [\text{bias}(\hat{\mu})]^2 + \text{var}(\hat{\mu}), \quad \text{where } \text{bias}(\hat{\mu}) := \mu - \mathbb{E}(\hat{\mu})\end{aligned}$$

IID sample means have **zero bias** and positive variance

Deterministic **LD** sample means have positive bias but **zero variance**



Shrinkage Estimators

Suppose that $\hat{\mu}_n$ is the sample mean of $Y_1, \dots, Y_n \stackrel{\text{IID}}{\sim} (\mu, \sigma^2)$.

For what value of α is $\alpha \hat{\mu}_n$ the estimator of μ with **smallest mean squared error**?



Acceptance-rejection sampling

Suppose that

- You **can** sample a random variable Z with known **proposal** density q_Z
- You really **want to** sample X with known (unnormalized) **target** density q , i.e., the true density is cq for some positive constant c
- And we **know** a M for which $q(x) \leq Mq_Z(x)$ for all x

The acceptance-rejection sampling proceeds as follows

```
i ← 0
repeat
  repeat
    generate  $Z \sim q_Z$  and  $U \sim \mathcal{U}[0,1]$ 
  until  $U \leq q(Z) / [Mq_Z(Z)]$ 
   $X[i] \leftarrow Z$ ,  $i \leftarrow i+1$ 
until  $i=n$ 
```




Why does acceptance–rejection sampling work?

$$W := \begin{cases} 1, & U \leq \varrho(Z)/[M\varrho_Z(Z)] \\ 0, & \text{otherwise} \end{cases} \quad \begin{array}{l} \text{accept} \\ \text{reject} \end{array}$$

Our sampling method generates X that has a PDF of

$$\begin{aligned} \tilde{\varrho}(z) &= \varrho_{Z|W}(z|1) = \text{PDF of } Z \text{ conditioned on accepting } Z \\ &= \frac{\varrho_{W|Z}(1|z)\varrho_Z(z)}{\varrho_W(1)} && \text{by Bayes Theorem} \\ &= \frac{\{\varrho(z)/[M\varrho_Z(z)]\}\varrho_Z(z)}{\varrho_W(1)} && \text{by our acceptance rule} \\ &= \frac{\varrho(z)}{M\varrho_W(1)} = c\varrho(z) \text{ since } c\varrho \text{ is a density; } \mathbb{P}(\text{acceptance}) = \varrho_W(1) = 1/(Mc) \end{aligned}$$

To get n samples of X , we need on average $n(Mc)$ samples of Z



Explorations in acceptance-rejection

See [Acceptance Rejection Sampling](#) for some examples and figures



Comparing Methods for Non-Uniform Sampling

Method	Requirements	LD friendly
$X = Q(U)$	quantile function	Yes
$\mathbf{X} = \mathbf{A}\mathbf{Z} + \boldsymbol{\mu} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$	$\Sigma = \mathbf{A}\mathbf{A}^T$	Yes
acceptance-rejection	$q(z) \leq M q_Z(z)$	Somewhat

For acceptance-rejection with LD see [\[Zhu & Dick 2024\]](#)
This is a good project topic